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Copulas

Syllabus objectives

1.3 Introduction to copulas

- 1.3.1 Describe how a copula can be characterised as a multivariate distribution function which is a function of the marginal distribution functions of its variates, and explain how this allows the marginal distributions to be investigated separately from the dependency between them.
- 1.3.2 Explain the meaning of the terms dependence or concordance, upper and lower tail dependence; and state in general terms how tail dependence can be used to help select a copula suitable for modelling particular types of risk.
- 1.3.3 Describe the form and characteristics of the Gaussian copula and the Archimedean family of copulas.

0 Introduction

Insurance and investment companies are often interested in being able to compute the joint probability of events occurring, for example the joint probability of losses on different classes of business or on investments, or the joint probability of default on investments. One way of calculating a joint probability is to use a joint PDF (or probability function in the case of discrete random variables) and then to integrate (or sum) this to find the probability.

There are a number of drawbacks to this approach. First of all, we would need to specify fully the joint distribution. This is not usually easy to determine unless, for example, all the underlying individual distributions are normal. Even if we can specify the joint distribution, it is not typically clear how the joint PDF (or probability function) relates to the individual PDFs (or probability functions) and what the nature of the association is between them.

An alternative way of calculating a joint probability is to use a copula. A copula is a function that takes as inputs marginal CDFs and outputs a joint CDF.

For example, suppose an insurer wants to work out the joint probability that annual losses on its household portfolio, X_H , will be less than or equal to £5m and that annual losses on its motor portfolio, X_M , will be less than or equal to £3m. For simplicity of calculation, we assume that the two portfolios give rise to losses independently.

Measuring in £m and using the assumption of independence, we calculate the probability as:

$$P(X_H \leq 5, X_M \leq 3) = P(X_H \leq 5)P(X_M \leq 3)$$

This calculation involves a copula function. We have taken the marginal CDFs as inputs:

$$F_{X_H}(5) = P(X_H \leq 5) \text{ and } F_{X_M}(3) = P(X_M \leq 3)$$

and outputted a joint CDF:

$$F_{X_H, X_M}(5, 3) = P(X_H \leq 5, X_M \leq 3)$$

The copula function that we've used here is called the *independence (or product) copula* and can be expressed as follows:

$$C[F_{X_H}(x_H), F_{X_M}(x_M)] = F_{X_H}(x_H) \times F_{X_M}(x_M) \text{ or as } C[u, v] = uv$$

where $u = F_{X_H}(x_H)$ and $v = F_{X_M}(x_M)$

Of course, it is not always the case that the relationship between the random variables is one of independence. However, there are many distinct copula functions, each of which expresses different types and levels of association between the variables.

In Sections 1 to 3 of this chapter we develop ideas leading to the definition of copula. In Sections 4 to 6 we look at different types of copula functions. In Sections 7 and 8 we introduce some practical applications of copulas.

1 Marginal and joint distributions

1.1 Joint distribution and density functions

The meaning of jointly distributed random variables, marginal distributions and conditional distributions was described in Subject CS1.

Recall from Subject CS1 that for two variables, the joint (cumulative) distribution function (CDF) is:

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$$

This can be extended from the bivariate case to the multivariate case in d dimensions:

$$F_{X_1, X_2, \dots, X_d}(x_1, x_2, \dots, x_d) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_d \leq x_d)$$

In the context of joint distribution functions, the individual distribution of each of the variables in isolation is known as its *marginal distribution*.

In the bivariate case, where random variables are continuous, we can derive the joint CDF by integrating the joint PDF, as follows:

$$F_{X,Y}(x,y) = \int_{s \leq x} \int_{t \leq y} f_{X,Y}(s,t) ds dt$$

1.2 Marginal distribution and density functions

To compute the marginal PDF from a joint PDF, we 'integrate out' the other variable:

$$f_X(x) = \int_y f_{X,Y}(x,y) dy$$

To compute a marginal CDF, we integrate the marginal PDF as follows:

$$F_X(x) = \int_{s \leq x} f_X(s) ds$$

The formulae above can be generalised from 2-dimensions (bivariate or joint case) to higher dimensions (multivariate case).

1.3 Standard joint distribution and density functions

Example ('off the shelf') multivariate statistical distributions are:

- the *multivariate normal distribution*, where the marginal distributions are all normal
- the *multivariate Student's t distribution*, where the marginal distributions are all t distributions.

The question below acts as a refresher on joint and marginal CDFs and PDFs and how they relate to each other.



Question

The joint PDF for two continuous random variables X and Y is:

$$f_{X,Y}(x,y) = \frac{1}{20}(x+4y), \quad 0 < x \leq 2, \quad 0 < y \leq 2$$

- (i) Derive a formula for the joint CDF, $F_{X,Y}(x,y)$.
- (ii) Derive formulae for the marginal PDFs, $f_X(x)$ and $f_Y(y)$, and comment on whether X and Y are independent.
- (iii) Derive formulae for the marginal CDFs, $F_X(x)$ and $F_Y(y)$.

Solution

(i) **Joint CDF**

The joint CDF is obtained by integrating the joint PDF with respect to both variables:

$$\begin{aligned}
 F_{X,Y}(x,y) &= \int_{s=0}^x \int_{t=0}^y f(s,t) \, ds \, dt \\
 &= \int_{s=0}^x \left(\int_{t=0}^y \frac{1}{20}(s+4t) \, dt \right) ds \\
 &= \frac{1}{20} \int_{s=0}^x \left[st + 2t^2 \right]_0^y ds \\
 &= \frac{1}{20} \int_{s=0}^x (sy + 2y^2) \, ds \\
 &= \frac{1}{20} \left[\left(\frac{1}{2}s^2 y + 2sy^2 \right) \right]_0^x \\
 &= \frac{1}{20} \left(\frac{1}{2}x^2 y + 2xy^2 \right) \\
 &= \frac{1}{40} xy(x+4y)
 \end{aligned}$$

(ii) **Marginal PDFs**

The marginal PDFs are obtained by ‘integrating out’ the other variable in the joint PDF:

$$f_X(x) = \int_{y=0}^2 f_{X,Y}(x,y) dy = \int_{y=0}^2 \frac{1}{20}(x+4y) dy = \frac{1}{20} \left[xy + 2y^2 \right]_0^2 = \frac{1}{20}(2x+8) = \frac{1}{10}(x+4)$$

$$f_Y(y) = \int_{x=0}^2 f_{X,Y}(x,y) dx = \int_{x=0}^2 \frac{1}{20}(x+4y) dx = \frac{1}{20} \left[\frac{1}{2}x^2 + 4xy \right]_0^2 = \frac{1}{20}(2+8y) = \frac{1}{10}(1+4y)$$

For X and Y to be independent, we must have:

$$f_{X,Y}(x,y) = f_X(x) \times f_Y(y)$$

In this case, the product of the marginal PDFs in (ii) is not equal to the joint PDF in (i). We therefore deduce that X and Y are *not* independent.

(iii) **Marginal CDFs**

The marginal CDFs are obtained by integrating the marginal PDFs:

$$F_X(x) = \int_0^x f_X(s) ds = \int_0^x \frac{1}{10}(s+4) ds = \frac{1}{10} \left[\frac{1}{2}s^2 + 4s \right]_0^x = \frac{1}{10} \left(\frac{1}{2}x^2 + 4x \right) = \frac{1}{20}x(x+8)$$

$$F_Y(y) = \int_0^y f_Y(t) dt = \int_0^y \frac{1}{10}(1+4t) dt = \frac{1}{10} \left[t + 2t^2 \right]_0^y = \frac{1}{10} (y + 2y^2) = \frac{1}{10} y(1+2y)$$

In the above question we have shown that the two random variables are not independent. However, the nature of the association is only *implicit* in the formulae for the joint PDF and joint CDF. We can’t immediately see the nature and extent of their association just by looking at the formula for the joint CDF.

Later in this chapter we will see how copula functions *explicitly* describe the *full* nature and extent of the association between random variables. However, before we do that let’s look at some simpler statistical measures of association.

2 Association, concordance, correlation and tail dependence

2.1 Introduction to terminology

Variables are said to be *associated* if there is some form of statistical relationship between them – whether causal or not. To facilitate comparisons, measures of association can be constructed.

Coefficients of association are generally designed so that their values vary between -1 and $+1$. Their absolute values increase with the strength of the relationship. They take a value of $+1$ (or -1) when there is perfect positive (or negative) association.

Any one particular type of coefficient of association measures a particular *form* of association. For example, Pearson's correlation coefficient measures the degree to which there is a *linear* relationship between the variables.

Concordance is another particular form of association. Broadly speaking, two random variables are concordant if small values of one are likely to be associated with small values of the other, and *vice versa*.

Spearman's rho and Kendall's tau (discussed in Subject CS1) are two examples of measures of concordance.

Note that a positive association between two variables does not necessarily imply that one is *dependent* on the other. For example, both might be dependent on a third (perhaps unobserved) variable, with neither being directly dependent on the other. A common pitfall for journalists is to forget that 'correlation does not imply causation'.

2.2 Pearson's linear correlation coefficient

We have previously met the linear correlation coefficient (also known as Pearson's ρ), which measures how strongly the values of two variables are related.

Pearson's rho is defined as:

$$\rho_{X,Y} = \frac{\text{cov}(X,Y)}{\sqrt{\text{var}(X)\text{var}(Y)}}$$



Question

Pearson's rho takes values between -1 and $+1$. Outline what it means when:

- (a) $\rho_{X,Y} = +1$
- (b) $\rho_{X,Y} = -1$
- (c) $\rho_{X,Y} = 0$.

Solution

- (a) When Pearson's rho is equal to +1, the random variables X and Y are perfectly positively correlated. This means that there is a perfectly increasing linear relationship between the values that the two random variables take.
- (b) When Pearson's rho is equal to -1, the random variables X and Y are perfectly negatively correlated. This means that there is a perfectly decreasing linear relationship between the values that the two random variables take.
- (c) When Pearson's rho is equal to 0, the random variables X and Y are uncorrelated. This means that there is no *linear* relationship between the values that the two random variables take.
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Since Pearson's rho is a measure of linear association between variables, it remains unchanged under a linear transformation of the variables, *ie* Pearson's rho for the random variables $aX + b$ and $cY + d$ has the same value as Pearson's rho for X and Y .

However, if we apply a non-linear transformation to the random variables, it is not necessarily the case that the value of Pearson's rho will stay the same. For example, if we calculate Pearson's rho for the random variables e^X and e^Y , this will not necessarily give us the same value as Pearson's rho for X and Y . In many respects this is counter-intuitive: we have applied a monotonically increasing function to the random variables and hence we might expect the degree of association to stay the same.

Desirable properties of a measure of concordance / association

A good measure of the *concordance* (or *association*) between two variables should have a number of properties. These include *invariance*, which requires that the measure of concordance does not change if we apply the same monotone function to the value of each variable. Pearson's ρ does not have this property.

2.3 Rank correlation coefficients

Two commonly used measures of concordance that are more robust than Pearson's ρ and are invariant are Spearman's ρ (often called the *rank correlation*) and Kendall's τ .

Measures of rank correlation look at the association between the position (or rank) of observations in a data series when they are arranged in order. For example, recall from Subject CS1, that Spearman's $s\rho$ can be calculated as:

$$s\rho = 1 - \frac{6}{T(T^2 - 1)} \sum_{i=1}^T d_i^2$$

where:

- T is the number of (pairs of) observations
- d_i is the difference in rank for the i th observation (pair).

Properties of rank correlation coefficients

Spearman's rho and Kendall's tau measure the degree of concordance between the rank (or position) rather than between the actual observations. This means that, even if the value of an observation changes, as long as its relative ranking stays the same, then the measure of concordance will remain the same. For example, we could apply an exponential transformation to the observations and both Spearman's rho and Kendall's tau would be unchanged.

2.4 Tail dependence

The correlation measures described above each try to 'summarise' the nature and extent of the association between variables into a single statistic. Two key shortcomings of such statistics are:

- information is lost in the summarisation process
- they capture the interdependency through the *whole* distribution, and this may be of less interest than the interdependency in the *tails* of the distribution.

It is often the case in insurance and investment applications that large losses tend to occur together. For example a hurricane could result in large losses on several different property insurance policies sold by the same company or a stock market crash could lead to large losses on a number of investments in the same investment portfolio.

So the relationships between the variables at the extremes of the distributions, ie in the upper and lower tails, are of particular importance. These can be measured using the coefficients of upper and lower tail dependence.



Coefficient of upper tail dependence

We can define the coefficient of upper tail dependence as:

$$\lambda_U = \lim_{u \rightarrow 1^-} P(X > F_X^{-1}(u) \mid Y > F_Y^{-1}(u))$$

This coefficient is a probability, so it takes a value between 0 and 1.

The coefficient of upper tail dependence indicates whether high values of one random variable, X , tend to be linked with high values of another random variable Y .

It considers the probability of the random variable X taking a value in the upper tail of its distribution (eg a tail with a probability mass of 5% $\Rightarrow u = 0.95$), given that the random variable Y takes a value in the same sized upper tail of its distribution.

Specifically, the coefficient of upper tail dependence is the limiting value of this probability as $u \rightarrow 1^-$, ie as we move further into the upper tail (from below).



Coefficient of lower tail dependence

The coefficient of lower tail dependence is defined as:

$$\lambda_L = \lim_{u \rightarrow 0^+} P(X \leq F_X^{-1}(u) \mid Y \leq F_Y^{-1}(u))$$

Again this coefficient is a probability, so takes a value between 0 and 1.

The coefficient of lower tail dependence indicates whether low values of one random variable, X , tend to be linked with low values of another random variable Y .

It considers the probability of the random variable X taking a value in the lower tail of its distribution (eg a tail with a probability mass of 5% $\Rightarrow u = 0.05$), given that the random variable Y takes a value in the same sized lower tail of its distribution.

Specifically, the coefficient of lower tail dependence is the limiting value of this probability as $u \rightarrow 0^+$, ie as we move further into the lower tail (from above).

3 Copulas

3.1 Expressing the association between variables explicitly

The joint distribution combines the information from the marginal distributions and the way in which the variables depend on each another. However, it expresses this dependence *implicitly*. We cannot immediately see the nature of the interdependence simply by looking at the formula for the joint distribution function.

Copulas provide an alternative approach that expresses the interdependence between the variables *explicitly*. They allow us to deconstruct the joint distribution of a set of variables into components (the marginal distributions plus a copula) that can be adjusted individually.

3.2 Definition of a copula

A **copula** is a function that expresses a multivariate cumulative distribution function in terms of the individual marginal cumulative distributions.

It is important to remember that a copula is a *function*. It takes marginal probabilities of random variables as inputs, and outputs a corresponding joint probability.



Definition of a copula

For a bivariate distribution, the copula is a function C_{XY} defined by:

$$C_{XY}[F_X(x), F_Y(y)] = P(X \leq x, Y \leq y) = F_{X,Y}(x, y)$$

This is often written in the more compact form:

$$C[u, v] = F_{X,Y}(x, y) \text{ where } u = F_X(x) \text{ and } v = F_Y(y)$$

This definition can be extended to the multivariate case where we have:

$$C[u_1, u_2, \dots, u_d] = F_{X_1, X_2, \dots, X_d}(x_1, x_2, \dots, x_d) \text{ where } u_i = F_{X_i}(x_i)$$

Note that the arguments $u_1, u_2, \dots, u_d$ and the output value of the copula function are restricted to the range $[0, 1]$, as they correspond to probabilities.

3.3 Three properties of copulas

Copulas must also satisfy three technical properties to ensure that they correctly capture the properties we would expect of a joint distribution in all circumstances.

Although not included in the Core Reading, we outline these additional three properties below:

1. A copula is an increasing function of its inputs:

$$C[u_1, \dots, u_i^*, \dots, u_d] > C[u_1, \dots, u_i, \dots, u_d] \text{ for } u_i^* > u_i \text{ and } i = 1, \dots, d$$

This makes sense from a probabilistic perspective because, if $u_i^* > u_i$, then

$P(X_i \leq x_i^*) > P(X_i \leq x_i)$ for corresponding $x_i^* = F_{X_i}^{-1}(u_i^*)$ and $x_i = F_{X_i}^{-1}(u_i)$, and hence:

$$P(X_1 \leq x_1, \dots, X_i \leq x_i^*, \dots, X_d \leq x_d) > P(X_1 \leq x_1, \dots, X_i \leq x_i, \dots, X_d \leq x_d)$$

2. If all the marginal CDFs are equal to 1 except for one of them, then the copula function is equal to the value of that one marginal CDF:

$$C(1, \dots, 1, u_i, 1, \dots, 1) = u_i \text{ for } i = 1, 2, \dots, d \text{ and } u_i \in [0, 1]$$

This makes sense because $u_k = 1 \Rightarrow P(X_k \leq x_k) = 1$ (ie a certainty), for $x_k = F_{X_k}^{-1}(u_k)$, and the only uncertainty in the above joint probability is the marginal probability with respect to the i th variable.

3. A copula function always returns a valid probability:

$$C[u_1, u_2, \dots, u_d] \in [0, 1]$$

3.4 Sklar's theorem

Sklar demonstrated in 1959 that the dependence structure of a set of random variables can be captured using copulas. The theorem is as follows:



Sklar's theorem

Let F be a joint (cumulative) distribution function with marginal cumulative distribution functions $F_1, \dots, F_d$. Then there exists a copula, C , such that for all $x_1, \dots, x_d \in [-\infty, \infty]$:

$$F(x_1, \dots, x_d) = C[F_1(x_1), \dots, F_d(x_d)]$$

In the case of variables that have a *continuous* distribution, the copula is unique.

Sklar's theorem tells us that if we have a joint CDF and marginal CDFs, then these can be linked by a copula function, ie a copula function exists.

The converse also holds:



Converse of Sklar's theorem

If C is a copula and $F_1, \dots, F_d$ are univariate cumulative distribution functions, then the function F defined above is a joint cumulative distribution function with marginal cumulative distribution functions $F_1, \dots, F_d$.



Question

The joint probability density function for two continuous random variables X and Y is:

$$f_{X,Y}(x,y) = \frac{1}{20}(x+4y), \quad 0 < x \leq 2, \quad 0 < y \leq 2$$

- (i) Derive formulae for the inverse cumulative distribution functions $F_X^{-1}(u)$ and $F_Y^{-1}(v)$.
- (ii) Hence derive a formula for the copula function $C(u,v) = F_{X,Y}(x,y)$.

Solution

(i) Inverse CDFs

In the solution to the question at the end of Section 1.3, we showed that:

$$F_X(x) = \int_0^x f_X(s) ds = \int_0^x \frac{1}{10}(s+4) ds = \frac{1}{10} \left[\frac{1}{2}s^2 + 4s \right]_0^x = \frac{1}{10} \left(\frac{1}{2}x^2 + 4x \right) = \frac{1}{20}x(x+8)$$

$$F_Y(y) = \int_0^y f_Y(t) dt = \int_0^y \frac{1}{10}(1+4t) dt = \frac{1}{10} \left[t + 2t^2 \right]_0^y = \frac{1}{10} (y + 2y^2) = \frac{1}{10}y(1+2y)$$

To find the inverse CDFs, we need to invert these CDFs by making x and y the subject of the equations, which in this case are quadratic equations.

We have:

$$u = F_X(x) = \frac{1}{20}x(x+8)$$

$$\Rightarrow x^2 + 8x - 20u = 0$$

$$\Rightarrow x = \frac{-8 \pm \sqrt{8^2 - 4(1)(-20u)}}{2(1)} = \frac{-8 \pm \sqrt{64 + 80u}}{2} = -4 \pm \sqrt{16 + 20u}$$

Only the positive root will result in values of x in the range $0 < x \leq 2$. So:

$$x = F_X^{-1}(u) = -4 + \sqrt{16 + 20u}$$

Similarly:

$$v = F_Y(y) = \frac{1}{10}y(1+2y)$$

$$\Rightarrow y = F_Y^{-1}(v) = \frac{1}{4} \left(-1 + \sqrt{1 + 80v} \right)$$

(ii) **Copula function**

The copula function, which is defined on the range $0 \leq u, v \leq 1$, is then obtained by writing the joint CDF in terms of u and v :

$$\begin{aligned} C[u, v] &= F_{XY}(x, y) \\ &= \frac{1}{40} xy(x + 80y) \\ &= \frac{1}{160} (-4 + \sqrt{16 + 20u}) (-1 + \sqrt{1 + 80v}) (-24 + \sqrt{16 + 20u} + 20\sqrt{1 + 80v}) \end{aligned}$$

3.5 Expressions of tail dependence and the survival copula

Lower tail dependence

Recall that the coefficient of lower tail dependence is defined as:

$$\lambda_L = \lim_{u \rightarrow 0^+} P(X \leq F_X^{-1}(u) \mid Y \leq F_Y^{-1}(u))$$

The coefficient of lower tail dependence measures the limit of this probability as the lower tail becomes smaller and smaller, as u tends to 0 from above. We can write:

$$\lambda_L = \lim_{u \rightarrow 0^+} P(X \leq x \mid Y \leq y)$$

where x and y are the lower tail percentage points of the distributions of the random variables X and Y respectively:

$$P(X \leq x) = P(Y \leq y) = u$$

Using the definition of conditional probabilities, the equation for the coefficient of lower tail dependence becomes:

$$\lambda_L = \lim_{u \rightarrow 0^+} \frac{P(X \leq x, Y \leq y)}{P(Y \leq y)}$$

The expression in the numerator of this fraction is a joint (cumulative) distribution function, *ie* it is the copula function $C[u, u]$. The expression in the denominator is the probability that Y takes a value less than or equal to y , which we know is u . This leads to the following useful formula:



Coefficient of lower tail dependence in terms of the copula function

$$\lambda_L = \lim_{u \rightarrow 0^+} \frac{C[u, u]}{u}$$

***ie* the coefficient of lower tail dependence can be calculated directly from the copula function.**

The coefficient of lower tail dependence can take values between 0 (no dependence) and 1 (full dependence).

The survival copula

To define the upper tail dependence, we need to look at the opposite end of the marginal distributions. Associated with each copula function is a *survival copula* function (indicated with a bar), which is defined by:

$$\bar{C}(x, y) = P(X > x, Y > y) = \bar{C}[\bar{F}_X(x), \bar{F}_Y(y)]$$

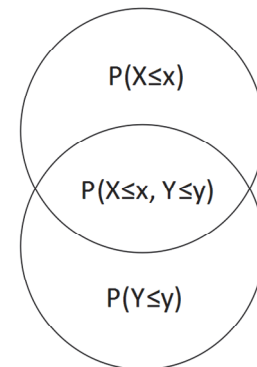
where $\bar{F}_X(x) = 1 - F_X(x)$ and $\bar{F}_Y(y) = 1 - F_Y(y)$.

By the principle of inclusion / exclusion, we have:

$$P(X \leq x \text{ and/or } Y \leq y) = P(X \leq x) + P(Y \leq y) - P(X \leq x, Y \leq y)$$

$$\text{ie } 1 - P(X > x, Y > y) = P(X \leq x) + P(Y \leq y) - P(X \leq x, Y \leq y)$$

$$\Rightarrow P(X > x, Y > y) = 1 - P(X \leq x) - P(Y \leq y) + P(X \leq x, Y \leq y)$$



So the survival copula is related to the original copula function by:

$$\bar{C}[1-u, 1-v] = 1 - u - v + C[u, v]$$



Coefficient of upper tail dependence in terms of the survival copula function

We can then define the coefficient of upper tail dependence as:

$$\lambda_U = \lim_{u \rightarrow 1^-} P(X > F_X^{-1}(u) \mid Y > F_Y^{-1}(u)) = \lim_{u \rightarrow 0^+} \frac{\bar{C}[u, u]}{u}$$

Let's look more closely at how the above result is derived.

The coefficient of upper tail dependence measures the limit of this probability as the upper tail becomes smaller and smaller, as u tends to 100% from below.

We have:

$$\lambda_U = \lim_{u \rightarrow 1^-} P(X > x \mid Y > y)$$

where x and y correspond to upper tail percentage points of the distributions of random variables X and Y respectively:

$$P(X \leq x) = P(Y \leq y) = u$$

and:

$$P(X > x) = P(Y > y) = 1 - u$$

Using the definition of conditional probabilities, the equation for the coefficient of upper tail dependence becomes:

$$\lambda_U = \lim_{u \rightarrow 1^-} \frac{P(X > x, Y > y)}{P(Y > y)}$$

The expression in the numerator of this fraction can be worked out using the principle of inclusion / exclusion result in the Core Reading above:

$$\begin{aligned} P(X > x, Y > y) &= 1 - P(X \leq x) - P(Y \leq y) + P(X \leq x, Y \leq y) \\ &= 1 - u - u + C[u, u] \\ &= 1 - 2u + C[u, u] \end{aligned}$$

The expression in the denominator is the probability that Y takes a value greater than y , which we know is $1 - u$. So we have the following formula:



Coefficient of upper tail dependence in terms of the copula function

$$\lambda_U = \lim_{u \rightarrow 1^-} \frac{1 - 2u + C[u, u]}{1 - u}$$

ie the coefficient of upper tail dependence can be calculated directly from the copula function.

Alternatively, we can express this using the survival copula function as:

$$\lambda_U = \lim_{u \rightarrow 1^-} \frac{\bar{C}[1-u, 1-u]}{1-u} = \lim_{u \rightarrow 0^+} \frac{\bar{C}[u, u]}{u}$$

3.6 Types of copula function

There are three main families of copula that we will go on to consider in the subsequent sections of this chapter:

- (i) fundamental copulas
- (ii) explicit copulas
- (iii) implicit copulas.

4 Fundamental copulas

Fundamental copulas represent the three basic (or fundamental) dependencies that a set of variables can display, namely:

- independence
- perfect positive interdependence, and
- perfect negative interdependence.

These copulas are referred to as the:

- independence (or product) copula
- co-monotonic (or minimum) copula
- counter-monotonic (or maximum) copula.

Collectively these three copulas are referred to as *fundamental copulas*. They are specific cases of a more general family of copulas called *Fréchet-Höfding copulas*.

In the bivariate case, the co-monotonic and counter-monotonic copulas represent the extremes of the possible levels of association between variables. They are therefore the upper and lower boundaries for all copulas – known as the *Fréchet-Höfding bounds*. The co-monotonic copula is the upper bound copula and the counter-monotonic copula is the lower bound copula.

4.1 Independence (or product) copula

One example of a bivariate copula is the *product copula* $C[u,v] = uv$. Here we have:

$$F_{X,Y}(x,y) = C[F_X(x), F_Y(y)] = F_X(x)F_Y(y)$$

or: $P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$

This captures the property of independence of the two variables X and Y , and so is also called the *independence (or product) copula*.

We looked at an example involving the independence (or product) copula in the introductory section of this chapter.



Question

Derive the coefficients of upper and lower tail dependence for the independence (or product) copula.

Solution

The independence (or product) copula is expressed as:

$$C[u,v] = uv$$

The coefficient of lower tail dependence is given by:

$$\begin{aligned}\lambda_L &= \lim_{u \rightarrow 0^+} \frac{C[u, u]}{u} \\ &= \lim_{u \rightarrow 0^+} \frac{u^2}{u} \\ &= \lim_{u \rightarrow 0^+} u \\ &= 0\end{aligned}$$

The coefficient of upper tail dependence is given by:

$$\begin{aligned}\lambda_U &= \lim_{u \rightarrow 1^-} \frac{1 - 2u + C[u, u]}{1 - u} \\ &= \lim_{u \rightarrow 1^-} \frac{1 - 2u + u^2}{1 - u} \\ &= \lim_{u \rightarrow 1^-} \frac{(1 - u)^2}{1 - u} \\ &= \lim_{u \rightarrow 1^-} (1 - u) \\ &= 0\end{aligned}$$

4.2 Co-monotonic (or minimum) copula

This copula is used where random variable demonstrate perfect positive interdependence. The *co-monotonic copula* is defined in the bivariate case as:

$$C[u, v] = \min(u, v)$$

Here we have:

$$C[F_X(x), F_Y(y)] = \min(F_X(x), F_Y(y))$$

or:
$$P(X \leq x, Y \leq y) = \min(P(X \leq x), P(Y \leq y))$$

To help us understand why the co-monotonic copula is the minimum of the two marginal CDFs, let's consider an example where two random variables X and Y display perfect positive interdependence.



Question

A statistician believes that there is a perfect positive interdependency between the Consumer Price Index (CPI) inflation rate, X (per annum), and the Retail Price Index (RPI) inflation rate, Y (per annum), and that the relationship can be modelled by the equation:

$$Y = X + 0.01$$

Show that $P(X \leq x, Y \leq y) = \min(P(X \leq x), P(Y \leq y))$.

Solution

We have:

$$\begin{aligned} P(X \leq x, Y \leq y) &= P(X \leq x, X + 0.01 \leq y) \\ &= P(X \leq x, X \leq y - 0.01) \end{aligned}$$

Now, for X to be less than or equal to both x and $y - 0.01$, it must be less than or equal to the smallest of these. So:

$$\begin{aligned} P(X \leq x, Y \leq y) &= P(X \leq \min(x, y - 0.01)) \\ &= \min(P(X \leq x), P(Y \leq y - 0.01)) \\ &= \min(P(X \leq x), P(Y - 0.01 \leq y - 0.01)) \\ &= \min(P(X \leq x), P(Y \leq y)) \end{aligned}$$

The second line uses the fact that $P(X \leq \min(a, b, c, \dots))$ is just the smallest of the probabilities $P(X \leq a), P(X \leq b), P(X \leq c), \dots$ ie $\min(P(X \leq a), P(X \leq b), P(X \leq c), \dots)$.

4.3 Counter-monotonic (or maximum) copula

The co-monotonic copula captures the relationship between two variables whose values are perfectly positively interdependent on each other, while the counter-monotonic copula captures the corresponding inverse relationship.

The *counter-monotonic copula* is defined in the bivariate case as:

$$C[u, v] = \max(u + v - 1, 0)$$

Here we have:

$$C[F_X(x), F_Y(y)] = \max(F_X(x) + F_Y(y) - 1, 0)$$

or: $P(X \leq x, Y \leq y) = \max(P(X \leq x) + P(Y \leq y) - 1, 0)$



Question

Let X and Y be two random variables that are perfectly negatively related as follows:

$$Y = -X$$

Show that $P(X \leq x, Y \leq y) = P(X \leq x) + P(Y \leq y) - 1$.

Solution

We have:

$$\begin{aligned} P(X \leq x, Y \leq y) &= P(X \leq x, -X \leq y) \\ &= P(X \leq x, X \geq -y) \\ &= P(-y \leq X \leq x) \\ &= P(X \leq x) - P(X < -y) \\ &= P(X \leq x) - P(-X > y) \\ &= P(X \leq x) - P(Y > y) \\ &= P(X \leq x) - [1 - P(Y \leq y)] \\ &= P(X \leq x) + P(Y \leq y) - 1 \end{aligned}$$

4.4 The multivariate case

The independence and co-monotonic copulas can be extended in the obvious way to the multivariate case. However, the counter-monotonic copula cannot. This is because it is not possible to have three or more variables where each pair has a direct inverse relationship.

In the multivariate case, we can extend the independence and co-monotonic copulas to d dimensions as follows:

$$\begin{aligned} \text{ind } C[F_{X_1}(x_1), \dots, F_{X_d}(x_d)] &= F_{X_1}(x_1) \times \dots \times F_{X_d}(x_d) \\ \text{min } C[F_{X_1}(x_1), \dots, F_{X_d}(x_d)] &= \min(F_{X_1}(x_1), \dots, F_{X_d}(x_d)) \end{aligned}$$

However, it is impossible to have three or more variables, *eg* X_1 , X_2 and X_3 , each of which always move in the *opposite* direction to all of the others. This is why the counter-monotonicity copula is only defined in two dimensions.

4.5 Graphical representation of copulas

Bivariate copulas can be represented graphically in various ways:

- scatterplots
- 3D (perspective) representations, and corresponding contour plots.

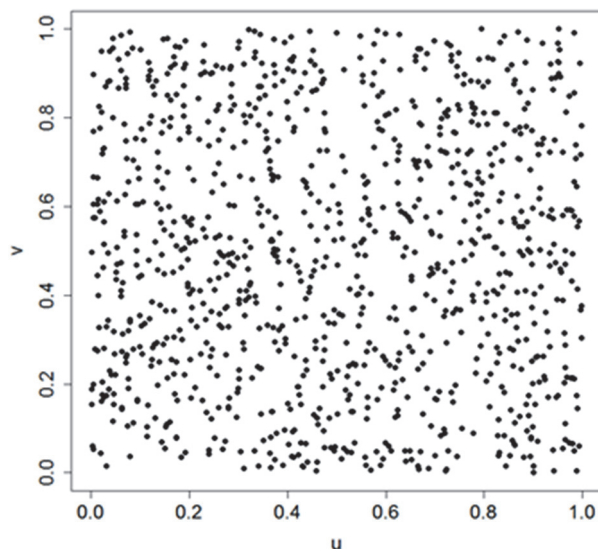
Scatterplots

The relationships implied by a copula can be illustrated using a scatterplot of simulated values of u and v .

Here we are recognising U and V as random variables, uniformly distributed on $[0,1]$, and illustrating the dependence structure between these two variables.

The diagrams below highlight the differences between the independence, co-monotonic and counter-monotonic copulas.

Scatterplot – Product (Independence) copula



This scatterplot of simulations illustrates the independence of the two variables.

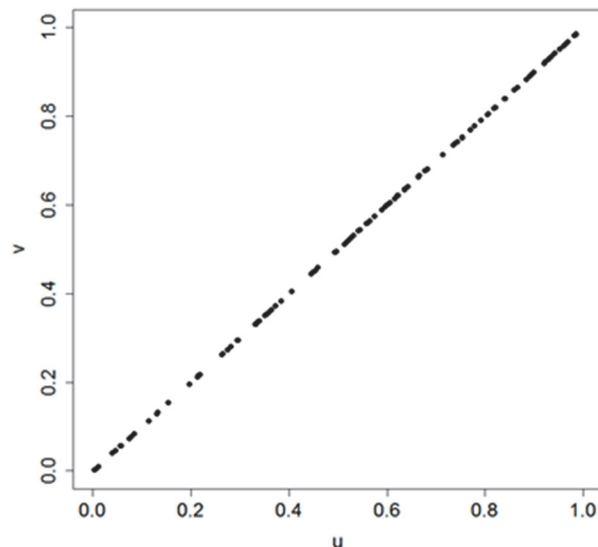
Each simulation (dot in the above diagram) has been created by:

- first simulating a particular value for U as a random number between 0 and 1
- then, independently, simulating the value for V as a random number between 0 and 1, because there is no association (dependence) between the two variables.

The variation in density of the dots in the above diagram is due to the random nature of the simulations, and the finite size of the sample. As the number of simulations increases the density will become ever more uniform.

In this case, if we know the marginal distributions of X and Y , and observe $X = x$, then we know that $P(X \leq x) = F_X(x) = u$. However, knowing this value of u tells us nothing about the value of v as there is no relationship between u and v .

Scatterplot – Co-monotonic copula



This scatterplot of simulations illustrates a perfect *positive* interdependence between the two variables.

Each simulation (dot in the above diagram) has been created by:

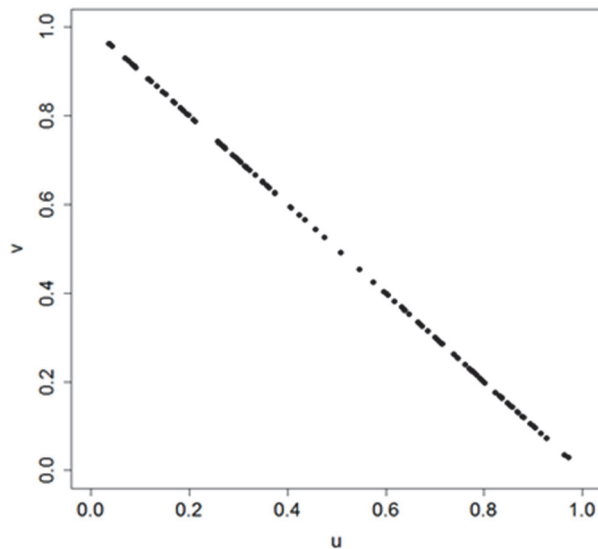
- first simulating a particular value for U as a random number between 0 and 1
- the value then taken by V is dictated by $u = v$.

As before, that the variation in density of the dots on the diagonal is due to the random nature of the simulations, and the finite size of the sample.

Since $x = F_X^{-1}(u)$ and $y = F_Y^{-1}(u)$, for all $0 \leq u \leq 1$, the diagram reinforces the fact that if X and Y are perfectly positively interdependent then:

- when X takes a value at a particular percentile of its distribution, Y also takes a value at the same percentile of its distribution
- when X takes a 'high' ('low') value in its possible range, Y also takes a 'high' ('low') value in its possible range.

Scatterplot – Counter-monotonic copula



This scatterplot of simulations illustrates a perfect *negative* interdependence between the two variables. Under such a relationship, given $U = u$ we can deduce that $V = v = 1 - u$.

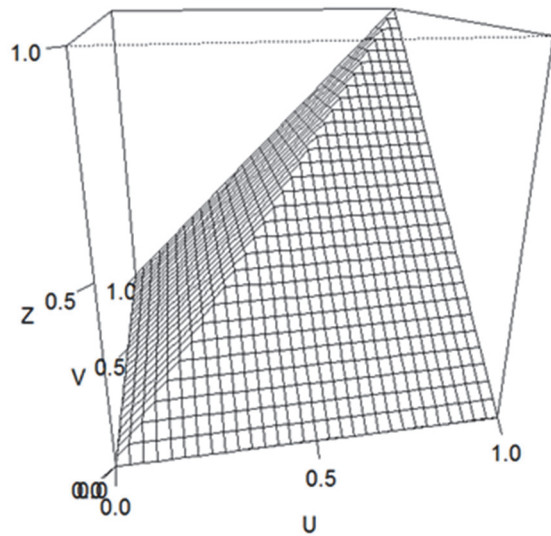
The diagram reinforces the fact that if X and Y are perfectly negatively interdependent, when X takes a 'high' ('low') value in its possible range, Y takes a 'low' ('high') value in its possible range.

These scatterplots are important, since, if a corresponding scatterplot based upon the underlying data shows features of independence, co-monotonicity or counter-monotonicity, it helps us to decide which copula function to fit to the data and, subsequently, use in our modelling.

3D representations and contour diagrams

The relationships described by copula functions, illustrated by the scatterplots (above), can also be represented in 3 dimensions: u , v and $C[u, v]$.

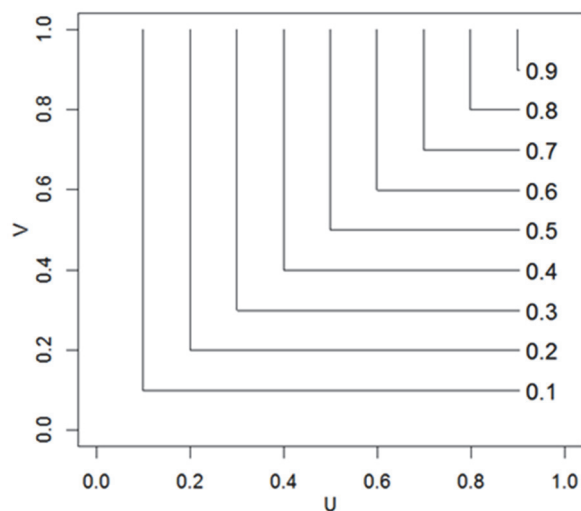
For example, the following two diagrams illustrate the co-monotonic copula. The first diagram shows the value of the copula plotted vertically in three dimensions. The second diagram shows contour lines of constant value of the copula function.



In general, we can interpret such a 3D-diagram as showing:

$$z = C[u, v] = P(X \leq x, Y \leq y)$$

where $u = P(X \leq x)$ and $v = P(Y \leq y)$. Note that $C[0,0] = 0$ and $C[1,1] = 1$. This is because the copula takes the form of a *cumulative* distribution function.



This second diagram is a two-dimensional representation of the first diagram. It is generally referred to as a contour diagram, as the (contour) lines indicate points of equal height in the 3D surface shown in the first diagram. (Think of taking a bird's eye view of the first diagram – looking down upon it from above.)

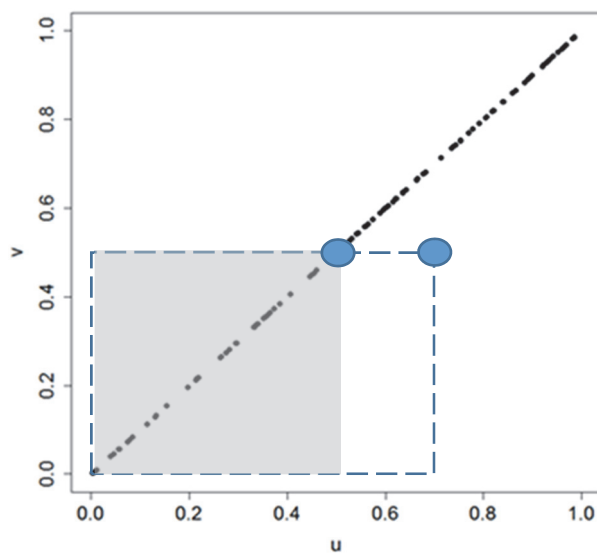
Question

By considering the scatterplot for the co-monotonic copula that was shown earlier, and recognising that $C[u, v] = P(U \leq u, V \leq v)$, explain why the second diagram (shown above) represents the co-monotonic copula.

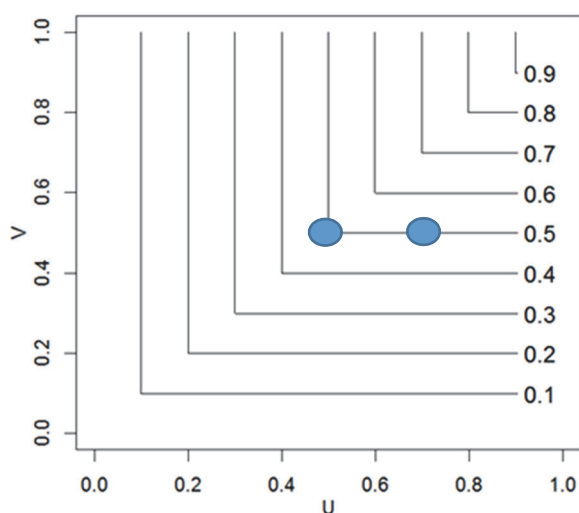
Solution

We can estimate $P(U \leq u, V \leq v)$ by looking at the number of simulations where $U \leq u$ and $V \leq v$ as a percentage of the total number of simulations. For example, in the simulation used to produce the scatterplot for the co-monotonic copula:

- Consider the condition $U \leq 0.5$ and $V \leq 0.5$, as indicated by the shaded square in the diagram below. About 50% of the simulations are where $U \leq 0.5$ and $V \leq 0.5$, ie the sample estimate of $P(U \leq 0.5, V \leq 0.5)$ is about 0.5.
- The same (number of) simulations meet the condition $U \leq 0.7$ and $V \leq 0.5$ (indicated by the dashed-line rectangle in the diagram below).

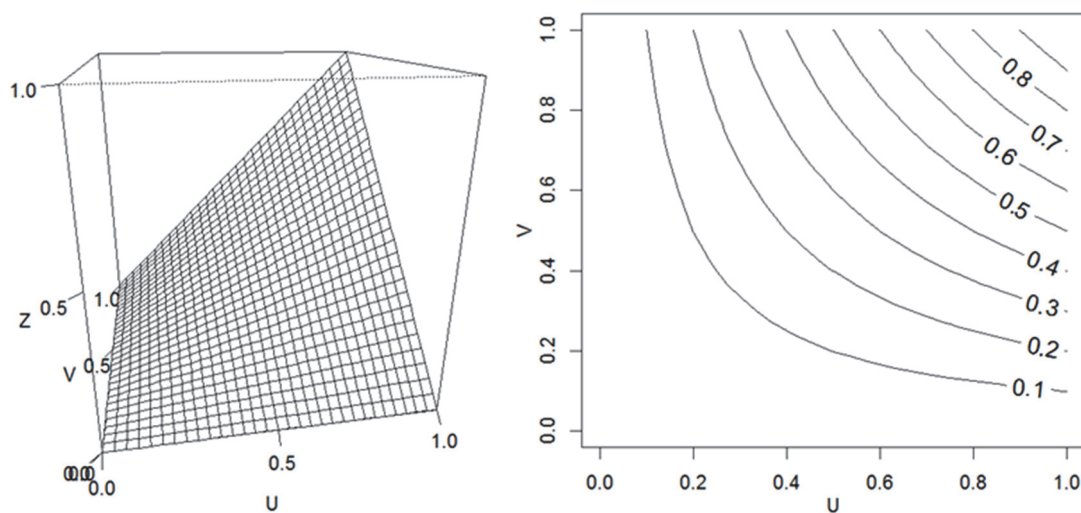


This is why (0.5,0.5) and (0.7,0.5) both lie on the same (0.5) contour line in the diagram below:

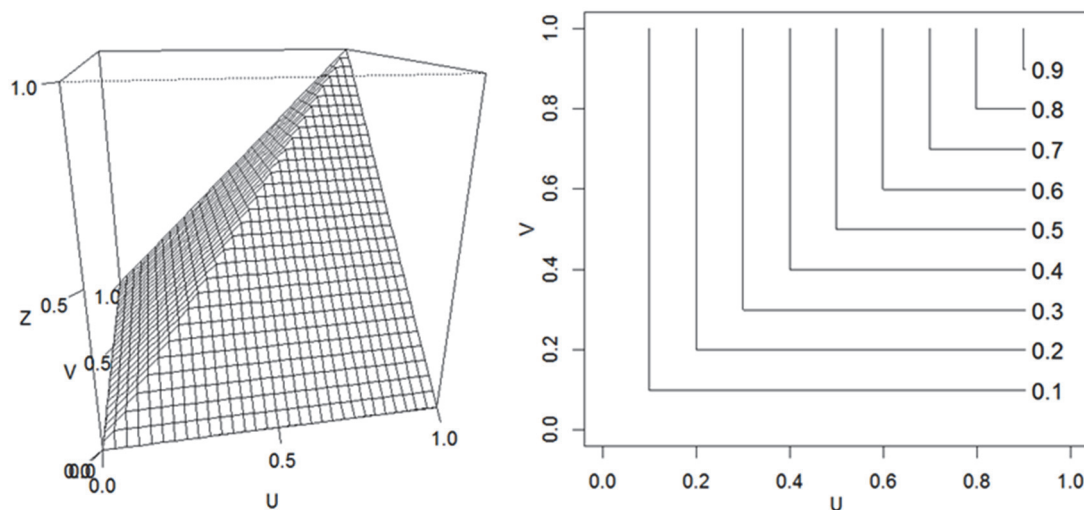


Similar perspective and contour diagrams (to those above) can be created for the other two fundamental copulas. All three sets of diagrams are shown on the next page for ease of comparison.

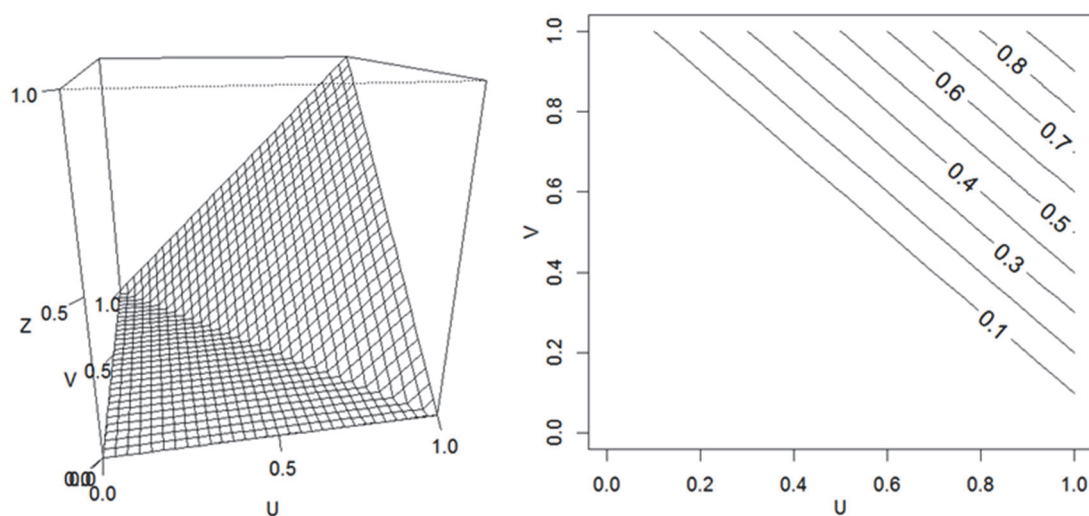
(i) independence (or product) copula:



(ii) co-monotonic (or minimum) copula



(ii) counter-monotonic (or maximum) copula:



5 Explicit copulas (including Archimedean copulas)

Explicit copulas have simple closed-form expressions. Below are three examples of commonly used explicit copulas.

As with statistical distributions such as the Poisson distribution or normal distribution, there are also a number of bespoke copulas that arise naturally in specific contexts.

Examples of these are:

- the Gumbel copula
- the Clayton copula
- the Frank copula.

In these examples, α is a parameter whose value can be specified. We will see later that this can be used to adjust the strength of the dependence between the variables.

5.1 Gumbel copula

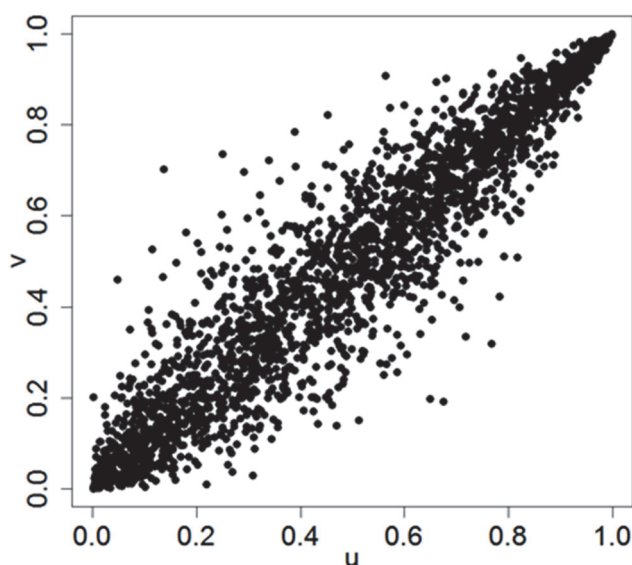
The *Gumbel copula* is defined in the bivariate case as:

$$C[u, v] = \exp \left\{ - \left((-\ln u)^\alpha + (-\ln v)^\alpha \right)^{1/\alpha} \right\}$$

Note that the Gumbel copula is often referred to as the Gumbel-Hougaard copula.

The Gumbel copula describes an interdependence structure in which there is upper tail dependence (the level of which is determined by the parameter α), but there is no lower tail dependence.

A scatterplot of simulated values from the Gumbel copula with parameter value $\alpha = 5$ is as follows:





Question

- (i) Derive the coefficient of upper tail dependence for the Gumbel copula.
- (ii) Comment on how the value of the parameter α affects the degree of upper tail dependence in the case of the Gumbel copula.
-

Solution

(i) ***Tail dependence of the Gumbel copula***

For the Gumbel copula setting $u = v$ gives:

$$\begin{aligned}
 C[u, u] &= \exp \left\{ - \left((-\ln u)^\alpha + (-\ln u)^\alpha \right)^{1/\alpha} \right\} \\
 &= \exp \left\{ - \left(2(-\ln u)^\alpha \right)^{1/\alpha} \right\} \\
 &= \exp \left\{ - \left(2^{1/\alpha} (-\ln u) \right) \right\} \\
 &= \exp \left\{ \left(2^{1/\alpha} \ln u \right) \right\} \\
 &= \exp \left\{ \ln u^{2^{1/\alpha}} \right\} \\
 &= u^{2^{1/\alpha}}
 \end{aligned}$$

The coefficient of upper tail dependence is given by:

$$\lambda_U = \lim_{u \rightarrow 1^-} \frac{1 - 2u + C[u, u]}{1 - u} = \lim_{u \rightarrow 1^-} \frac{1 - 2u + u^{2^{1/\alpha}}}{1 - u}$$

In the limit this fraction has the form $\frac{0}{0}$, which is undefined. However, we can use L'Hôpital's

rule, $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$, to find the value of the limit:

$$\lambda_U = \lim_{u \rightarrow 1^-} \frac{1 - 2u + u^{2^{1/\alpha}}}{1 - u} = \lim_{u \rightarrow 1^-} \frac{-2 + 2^{1/\alpha} u^{2^{1/\alpha} - 1}}{-1} = 2 - 2^{1/\alpha}$$

(ii) ***Comment***

As α increases, $2^{1/\alpha}$ reduces and hence $2 - 2^{1/\alpha}$ increases. So increasing the value of the parameter α increases the degree of upper tail dependence of the Gumbel copula.

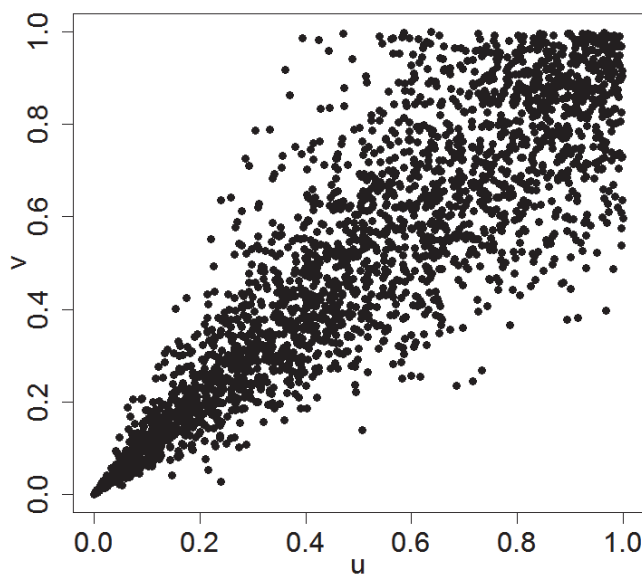
5.2 Clayton copula

The *Clayton copula* is defined in the bivariate case as:

$$C[u, v] = (u^{-\alpha} + v^{-\alpha} - 1)^{-1/\alpha}$$

The Clayton copula describes an interdependence structure in which there is lower tail dependence (the level of which is determined by the parameter α), but there is no upper tail dependence.

A scatterplot of simulated values from the Clayton copula with parameter value $\alpha = 5$ is as follows:



From the diagram:

- We might anticipate that there *may* be a degree of lower tail dependence because the simulations in a 'thin rectangle' at the bottom of the diagram lie mostly in the 'small square at the left-hand end' (ie if $V \leq v$, for small values of v , then it is likely that $U \leq v$).



- We might anticipate that there *may* be a lack of upper tail dependence because the simulations in a 'thin rectangle' at the top of the diagram do not lie mostly in the 'small square at the right-hand end' (ie if $V \geq v$, for large values of v , then we are not confident that $U \geq v$).



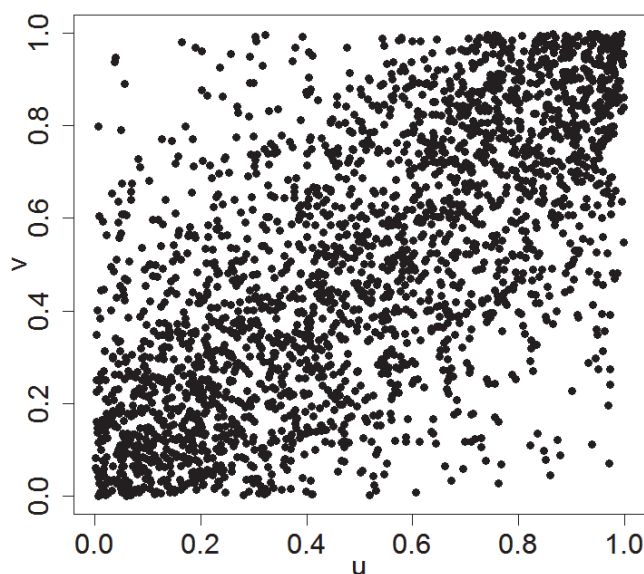
5.3 Frank copula

The *Frank copula* is defined in the bivariate case as:

$$C[u, v] = -\frac{1}{\alpha} \ln \left(1 + \frac{(e^{-\alpha u} - 1)(e^{-\alpha v} - 1)}{(e^{-\alpha} - 1)} \right)$$

The Frank copula describes an interdependence structure in which there is no upper or lower tail dependence.

A scatterplot of simulated values from the Frank copula with parameter value $\alpha = 5$ is as follows:



The lack of tail dependency is indicated in the scatterplot in that, for example, if V takes a value very close to 0 or 1, then we are uncertain whether or not U will also take a value similarly close to 0 or 1.

5.4 Archimedean copulas

A number of copulas can be specified by a special form of generator function that automatically captures the properties required for a copula. These are called *Archimedean copulas*.

Archimedean copulas are described by reference to a *generator function*. In the bivariate case, they take the form:

$$C[u, v] = \psi^{[-1]}(\psi(u) + \psi(v))$$

where $\psi(x)$ is the generator function, and $\psi^{[-1]}$ is the *pseudo-inverse function* (explained below).

Archimedean copulas are a subset of explicit copulas. The Gumbel, Clayton and Frank copulas are all examples of Archimedean copulas.

For example, consider the Gumbel copula:

$$C[u, v] = \exp \left\{ - \left((-\ln u)^\alpha + (-\ln v)^\alpha \right)^{1/\alpha} \right\}$$

We can see that the Gumbel copula is an example of an Archimedean copula where the generator function is $\psi(x) = (-\ln x)^\alpha$ and the inverse generator function is $\psi^{-1}(x) = \exp(-x^{1/\alpha})$.

Before we go any further with Archimedean copulas, we need to introduce some maths relating to pseudo-inverse functions.

Pseudo-inverse functions

In order to define Archimedean copulas, we need to extend the familiar idea of an inverse function (eg the inverse of $f(x) = x^2$ is $f^{-1}(x) = \sqrt{x}$) to ensure that the inverse function is defined for all possible arguments. This is done by defining the *pseudo-inverse* function $\psi^{[-1]}(x)$ of a function $\psi(x)$ as:

$$\psi^{[-1]}(x) = \begin{cases} \psi^{-1}(x) & \text{if } 0 \leq x \leq \psi(0) \\ 0 & \text{if } \psi(0) < x \leq \infty \end{cases}$$

where $\psi^{-1}(x)$ denotes the ordinary inverse function obtained by inverting the equation $x = \psi(y)$ to express y in terms of x .

The pseudo-inverse function gives us a means of determining the inverse where the function ψ outputs values on a finite rather than infinite range.

If $\psi(0) = \infty$, the pseudo-inverse is always equal to the 'ordinary' inverse and the generator function is called a *strict generator function*.

General definition of an Archimedean copula

The definition of Archimedean copulas can be extended to more than 2 dimensions.

Copulas in the Archimedean family are of the form:

$$C[u_1, \dots, u_d] = \psi^{[-1]} \left(\sum_{i=1}^d \psi(u_i) \right)$$

In order to be valid, the generator function $\psi: [0, 1] \rightarrow [0, \infty]$ must be a continuous, strictly decreasing, convex function with $\psi(1) = 0$.

Although this may all seem a little theoretical and complicated, the idea behind Archimedean copulas is, in fact, very intuitive. Remember that a copula function takes as inputs marginal CDFs and outputs a joint CDF. Using the generator function and its inverse, we are:

- taking probabilities between 0 and 1 (the u_i 's or marginal CDFs)
- converting these to numbers greater than 0 using the generator function ψ
- summing the results
- converting the result back to a probability (ie the joint CDF) using the inverse function ψ^{-1} .

Note that the definition of the pseudo-inverse function ensures that, whatever the sum

$\sum_{i=1}^d \psi(u_i)$, the value of the Archimedean copula will always be a valid probability.

In the bivariate case, we have:

$$C[u, v] = \psi^{[-1]}(\psi(u) + \psi(v))$$

We now look at how the Gumbel, Clayton and Frank copulas can be derived using the generator function approach.

Gumbel copula

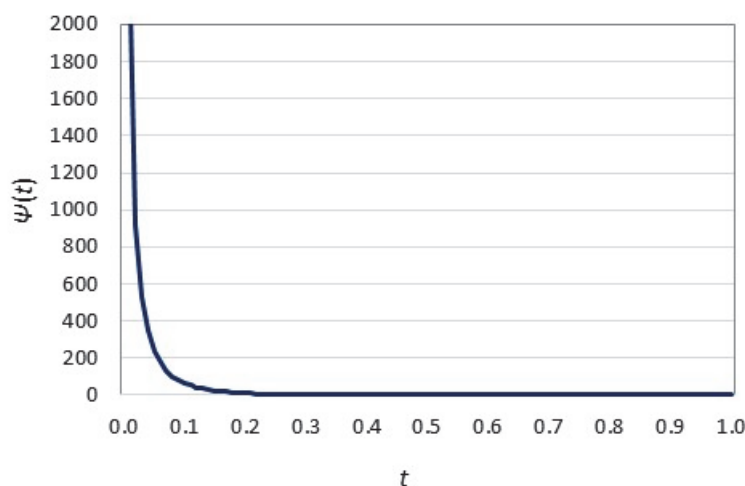
For example, the *Gumbel copula* can be defined by the generator function:

$$\psi(t) = (-\ln t)^\alpha \text{ where } 1 \leq \alpha < \infty$$

which we can use to deduce an explicit formula for the copula function.

In this case $\psi(0) = \lim_{t \rightarrow 0} (-\ln t)^\alpha = \infty$, so the pseudo-inverse function equals the normal inverse function.

The graph of the Gumbel generator function where $\alpha = 5$ is illustrated below:



We can see from this graph of the Gumbel generator function that it looks valid, ie it is a continuous, strictly decreasing, convex function, $\psi: [0,1] \rightarrow [0,\infty]$, with $\psi(1)=0$.

The inverse generator function can be found by inverting the relationship $x = (-\ln y)^\alpha$ to obtain $y = \exp(-x^{1/\alpha})$, so that $\psi^{[-1]}(x) = \psi^{-1}(x) = \exp(-x^{1/\alpha})$. We then have:

$$C[u,v] = \psi^{[-1]}(\psi(u) + \psi(v)) = \exp\left\{-\left((- \ln u)^\alpha + (- \ln v)^\alpha\right)^{1/\alpha}\right\}$$



Question

Confirm algebraically that the Gumbel generator function, $\psi(t) = (-\ln t)^\alpha$, is valid.

Solution

The generator function is $\psi(t) = (-\ln t)^\alpha$.

- When $t=0$, $\psi(0) = \lim_{t \rightarrow 0} (-\ln t)^\alpha = \infty$.
- When $t=1$, $\psi(1) = (-\ln 1)^\alpha = 0$.
- In the range $0 < t < 1$, $\ln t$ takes increasingly negative values, so that $-\ln t$ takes decreasing positive values, and hence so does $\psi(t) = (-\ln t)^\alpha$.

Hence the generator function, $\psi(t) = (-\ln t)^\alpha$, for the Gumbel copula is valid.

Clayton copula

The *Clayton copula* is defined by the generator:

$$\psi(t) = \frac{1}{\alpha} (t^{-\alpha} - 1) \text{ where } -1 \leq \alpha < \infty$$

Frank copula

The *Frank copula* is defined by the generator:

$$\psi(t) = -\ln \left(\frac{e^{-\alpha t} - 1}{e^{-\alpha} - 1} \right) \text{ where } -\infty < \alpha < \infty$$

Independence (or product) copula

The independence (or product) copula is also Archimedean. Its generator is $\psi(t) = -\ln t$.



Question

Show that the generator function $\psi(t) = -\ln t$ generates the independence (or product) copula.

Solution

The inverse function is found by rearranging the equation $x = -\ln t$ to make t the subject of the formula:

$$x = \psi(t) = -\ln t \Rightarrow t = \psi^{-1}(x) = \exp(-x)$$

We substitute in the generator and inverse generator functions giving:

$$\begin{aligned} C[u,v] &= \psi^{-1}((- \ln u) + (- \ln v)) \\ &= \exp[-(- \ln uv)] \\ &= uv \end{aligned}$$

The generator function is valid because:

- $\psi(0) = \lim_{t \rightarrow 0} (-\ln t) = \infty$.
 - $\psi(1) = -\ln 1 = 0$.
 - In the range $0 < t < 1$, $\ln t$ takes increasing values, so that $-\ln t$ takes decreasing values.
-

6 Implicit copulas

The final group of copulas that we consider are called implicit copulas. These copulas are based on (or implied by) well-known multivariate distributions, but no simple closed-form expression exists for them. We look at:

- the Gaussian copula (based on the multivariate normal distribution)
- the Student's t copula (based on the multivariate Student's t distribution).

6.1 Gaussian copula

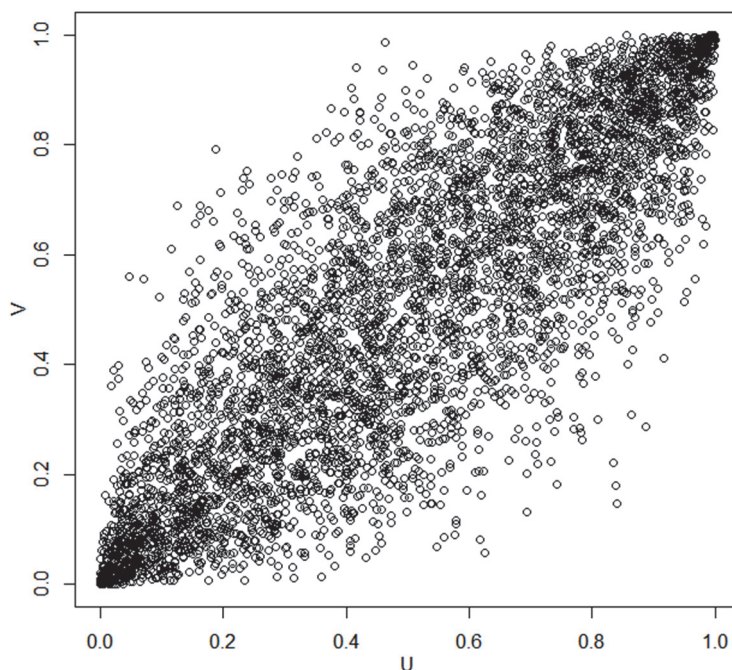
The bivariate *Gaussian copula* is defined by:

$$C[u, v] = \Phi_{\rho} \left[\Phi^{-1}(u), \Phi^{-1}(v) \right]$$

where Φ is the distribution function of the standard normal distribution and Φ_{ρ} is the distribution function of a bivariate normal distribution with correlation ρ .

Applying this Gaussian copula to normal marginal distributions will result in a bivariate normal distribution with correlation ρ .

An example scatterplot for the Gaussian copula, with a correlation parameter of $\rho = 0.85$ is:



In this example ($\rho = 0.85$):

- if $u = 1$, then it is extremely unlikely that we observe $v = 0$
- if $u = 0$, then it is extremely unlikely that we observe $v = 1$.

The independence, co-monotonic and counter-monotonic copulas are special cases of the Gaussian copula where $\rho=0$, $\rho=+1$ and $\rho=-1$, respectively.

Unless the correlation parameter is equal to 1, *ie* unless the two random variables are perfectly positively interdependent, there is no tail dependence (upper or lower) exhibited by the Gaussian copula.

The bivariate Gaussian copula can be extended to the multivariate case incorporating the $d \times d$ correlation matrix of the individual random variables. So this is the unique copula that reproduces a joint normal distribution with a specified correlation matrix from the individual marginal distributions. Because it reproduces the joint distribution in this way, it is sometimes called an *implicit* copula.

The formula defining the bivariate Gaussian copula is mathematically equivalent to the following integral form:

$$C[u, v] = \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\Phi^{-1}(u)} \int_{-\infty}^{\Phi^{-1}(v)} \exp\left\{-\frac{1}{2(1-\rho^2)}(s^2 + t^2 - 2\rho st)\right\} ds dt$$

This can be simplified further to:

$$C[u, v] = \int_0^u \Phi\left(\frac{\Phi^{-1}(v) - \rho\Phi^{-1}(t)}{\sqrt{1-\rho^2}}\right) dt$$

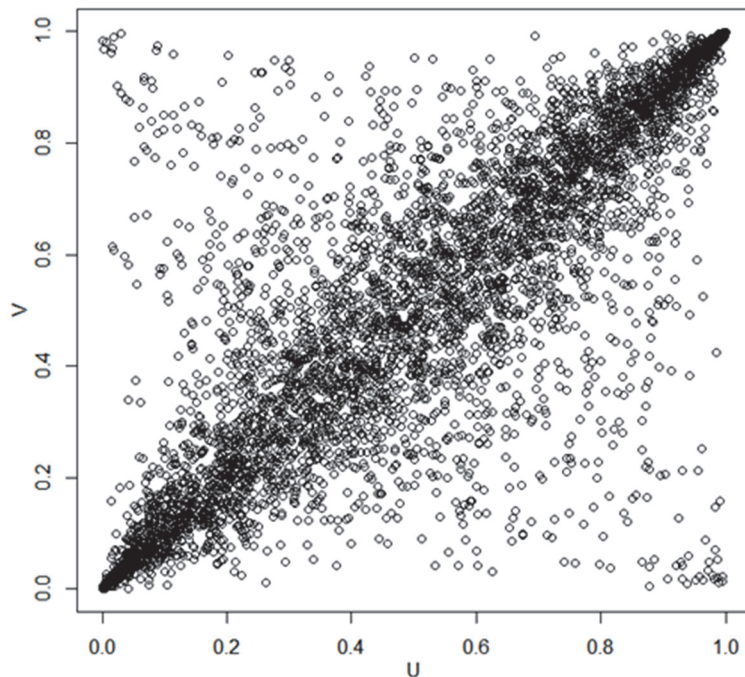
6.2 Student's t copula

The *Student's t copula* is defined by:

$$C[u, v] = t_{\gamma, \rho}\left[t_{\gamma}^{-1}(u), t_{\gamma}^{-1}(v)\right]$$

where t_{γ} is the distribution function of a random variable with a Student's t distribution with γ degrees of freedom and $t_{\gamma, \rho}$ is the distribution function of a bivariate Student's t distribution with correlation ρ .

An example scatterplot for the Student's t copula, with a correlation coefficient of $\rho=0.85$ and $\gamma=1$, is shown below.



There are some simulations in the upper-left and lower-right corners of this scatterplot, which was not the case in the example scatterplot for the Gaussian copula that we saw earlier (and which was based on the same correlation parameter).

The Student's t copula allows the dependencies between the variables to be adjusted more finely than the corresponding Gaussian copula.

One of the disadvantages of the Gaussian copula, is that it has just a single parameter, ρ . The Student's t copula has an additional parameter, γ , the number of degrees of freedom, which can be used to vary the strength of the association between the two variables, *ie* the degree of tail dependence. The smaller the value of γ , the greater the level of tail dependence.

In the same way that the standard normal distribution is a limiting case of the Student's t distribution (as the number of degrees of freedom tends to infinity), the Gaussian copula is the limiting case of the Student's t copula.

Specifically, as the number of degrees of freedom tends to infinity, $\gamma \rightarrow \infty$, the Student's t copula tends to approach the Gaussian copula.

7 Choosing and fitting a suitable copula function

7.1 Choosing a suitable copula function

If we want to create a mathematical model to represent real-world phenomena then we might look at past data and:

- select and parameterise marginal distributions for each of the relevant variables, and
- describe and quantify the form and extent of the associations between the variables.

Examination of the form and levels of association between the variables of interest allows us to select a suitable candidate copula from the list of established copulas or to develop a new bespoke copula.

Different copulas result in different levels of tail dependence.

For example:

- the Frank copula and the Gaussian copula have zero dependence in both tails, while the Student's t copula has equal positive dependence in both tails.
- the Gumbel copula has zero lower tail dependence but upper tail dependence of $2 - 2^{1/\alpha}$. The Clayton copula, on the other hand, has zero upper tail dependence but lower tail dependence of $2^{-1/\alpha}$.

As we would expect, variables related by the independence (product) copula have a concordance of 0, whereas variables related by the co-monotonic (minimum) or counter-monotonic (maximum) copulas have a concordance of +1 and -1 respectively.

We can summarise the upper and lower tail dependence results in the table below:

Copula name	$_L\lambda$	$_U\lambda$
Independence	0	
Co-monotonic	1	
Counter-monotonic	0	
Gumbel	0	$2 - 2^{1/\alpha}$
Clayton	$2^{-1/\alpha}$ if $\alpha > 0$ 0 if $\alpha \leq 0$	0
Frank	0	
Gaussian	0 if $\rho < 1$ 1 if $\rho = 1$	
Student's t	> 0 if $\gamma < \infty$, increasing as γ decreases 0 if $\gamma = \infty$ and $\rho \neq 1$ 1 for all γ when $\rho = 1$	

So the Gumbel copula, with an appropriate value for the parameter α , might be a suitable copula to use when modelling large general insurance claims resulting from a common underlying cause.



Question

State an appropriate copula to use if the data exhibit the following features:

- (a) independence
- (b) high upper tail dependence, but no lower tail dependence
- (c) a high degree of positive interdependence throughout
- (d) high lower tail dependence, but no upper tail dependence
- (e) a high degree of negative interdependence throughout
- (f) no upper or lower tail dependence
- (g) both upper and lower tail dependence.

Solution

- (a) the independence (or product) copula
 - (b) the Gumbel copula
 - (c) the co-monotonic copula (perfect positive interdependence), or Frank copula (with a high positive parameter to give a strong positive association throughout)
 - (d) the Clayton copula
 - (e) the counter-monotonic copula, (perfect negative interdependence), or Frank copula (with a high negative parameter to give a strong negative association throughout)
 - (f) the Frank copula, Gaussian copula, or independence copula
 - (g) the Student's t copula
-

7.2 Fitting a copula

This short section is beyond the syllabus of Subject CS2.

We have seen above how the form and level of association between the variables (including the degree of lower and upper tail dependence) can help us choose a particular copula. The final step is how to parameterise that chosen copula. For example, if we have chosen to use the Gumbel copula in our model, we then have to choose a suitable value of the parameter α .

For Archimedean copulas, Kendall's tau is a function of the parameters of the copula. Therefore, we can fit Gumbel, Clayton and Frank copulas using a method of moments approach. We would calculate Kendall's tau for the observations and equate this to the formula for Kendall's tau for the copula, solving to obtain the fitted value of α .

For example, for the Gumbel copula, Kendall's tau is given by the formula:

$$\tau = 1 - \frac{1}{\alpha}$$

If the observed value of Kendall's tau is 0.75, then using the method of moments gives:

$$\begin{aligned} 1 - \frac{1}{\hat{\alpha}} = 0.75 &\Rightarrow \frac{1}{\hat{\alpha}} = 0.25 \\ &\Rightarrow \hat{\alpha} = 4 \end{aligned}$$

For the Clayton copula, Kendall's tau is given by the formula:

$$\tau = \frac{\alpha}{\alpha + 2}$$

If the observed value of Kendall's tau is 0.75, then using the method of moments gives:

$$\begin{aligned} \frac{\hat{\alpha}}{\hat{\alpha} + 2} = 0.75 &\Rightarrow \hat{\alpha} = 0.75\hat{\alpha} + 1.5 \\ &\Rightarrow 0.25\hat{\alpha} = 1.5 \\ &\Rightarrow \hat{\alpha} = 6 \end{aligned}$$

A similar method can be used for the Frank copula, although the formula for Kendall's tau is more complicated.

It is also possible to use the method of maximum likelihood estimation to fit the copula parameters.

8 Calculating probabilities using copulas

Recall that a copula:

- is a function
- takes marginal probabilities of random variables as inputs, and outputs a corresponding joint probability
- for a bivariate distribution is defined by:

$$C_{XY}[F_X(x), F_Y(y)] = P(X \leq x, Y \leq y) = F_{X,Y}(x, y)$$

which is often written in the more compact form:

$$C[u, v] = F_{X,Y}(x, y) \text{ where } u = F_X(x) \text{ and } v = F_Y(y)$$



Question

Let X = a person's height measured in cm , and Y = weight measured in kg . Heights and weights are each assumed to be normally distributed, and:

$$P(X \leq 180) = 0.81594 \text{ and } P(Y \leq 70) = 0.69146$$

- (i) Calculate the joint probability that a person's height is less than or equal to $180cm$ and that their weight is less than or equal to $70kg$ using:
- the independence (or product) copula
 - the Gaussian copula with $\rho = 0$.

The following table is required for (i)(a). It shows an excerpt of values from the bivariate standard normal cumulative distribution function: $\Phi_{\rho=0}(x, y)$.

		X										
	$\Phi(x, y)$	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
	0	0.2500	0.2699	0.2896	0.3090	0.3277	0.3457	0.3629	0.3790	0.3941	0.4080	0.4207
	0.1	0.2699	0.2914	0.3127	0.3336	0.3538	0.3733	0.3918	0.4092	0.4255	0.4405	0.4542
	0.2	0.2896	0.3127	0.3355	0.3579	0.3797	0.4005	0.4204	0.4391	0.4565	0.4726	0.4874
	0.3	0.3090	0.3336	0.3579	0.3818	0.4050	0.4273	0.4484	0.4684	0.4870	0.5042	0.5199
	0.4	0.3277	0.3538	0.3797	0.4050	0.4296	0.4532	0.4757	0.4968	0.5166	0.5348	0.5514
Y	0.5	0.3457	0.3733	0.4005	0.4273	0.4532	0.4781	0.5018	0.5242	0.5450	0.5642	0.5818
	0.6	0.3629	0.3918	0.4204	0.4484	0.4757	0.5018	0.5267	0.5501	0.5720	0.5922	0.6106
	0.7	0.3790	0.4092	0.4391	0.4684	0.4968	0.5242	0.5501	0.5746	0.5974	0.6185	0.6378
	0.8	0.3941	0.4255	0.4565	0.4870	0.5166	0.5450	0.5720	0.5974	0.6212	0.6431	0.6631
	0.9	0.4080	0.4405	0.4726	0.5042	0.5348	0.5642	0.5922	0.6185	0.6431	0.6658	0.6865
	1	0.4207	0.4542	0.4874	0.5199	0.5514	0.5818	0.6106	0.6378	0.6631	0.6865	0.7079

- (ii) Compare the two results in (i).

Solution

We have $u = P(X \leq 180) = 0.81594$ and $v = P(Y \leq 70) = 0.69146$.

(i)(a) **Independence (or product) copula**

The independence (or product) copula is given by:

$$C[u, v] = uv$$

The joint probability that a person's height is less than or equal to 180cm and their weight is less than or equal to 70kg is given by the independence (or product) copula as:

$$P(X \leq 180, Y \leq 70) = 0.81594 \times 0.69146 = 0.5642$$

(i)(b) **Gaussian copula with perfect positive correlation**

The Gaussian copula is given by:

$$C[u, v] = \Phi_{\rho} \left[\Phi^{-1}(u), \Phi^{-1}(v) \right]$$

The joint probability that a person's height is less than or equal to 180cm and their weight is less than or equal to 70kg is given by the Gaussian copula with $\rho = 0$ as:

$$P(X \leq 180, Y \leq 70) = \Phi_{\rho} \left[\Phi^{-1}(0.81594), \Phi^{-1}(0.69146) \right]$$

From page 160 of the *Tables*, we have:

$$\Phi^{-1}(0.81594) = 0.90$$

and $\Phi^{-1}(0.69146) = 0.50$

Therefore:

$$P(X \leq 180, Y \leq 70) = \Phi_{\rho} [0.90, 0.50]$$

Using the excerpt provided in the question for the bivariate standard normal CDF:

$$P(X \leq 180, Y \leq 70) = \Phi_{\rho} [0.90, 0.50] = 0.5642$$

(ii) **Compare the results**

The results of both calculations are the same. This is because, for the multivariate Gaussian distribution, a correlation coefficient of 0 implies independence.

The chapter summary starts on the next page so that you can keep all the chapter summaries together for revision purposes.

Chapter 17 Summary

Association and concordance

Variables are said to be associated if there is some form of statistical relationship between them, whether causal or not.

Coefficients of association are generally designed so that their values vary between -1 and $+1$. Their absolute values increase with the strength of the relationship. They take a value of $+1$ (or -1) when there is perfect positive (or negative) association. A positive association between two variables does not necessarily imply that one is *dependent* on the other.

Pearson's correlation coefficient measures the degree to which there is a linear relationship between the variables.

Concordance is another particular form of association. Broadly speaking, two random variables are concordant if small values of one are likely to be associated with small values of the other, and vice versa.

Spearman's rho and Kendall's tau are two examples of measures of concordance.

Definition of a copula and properties

A copula function takes as its inputs the marginal cumulative distribution functions, and outputs a joint cumulative distribution function. A copula in d -dimensions is expressed as:

$$C[u_1, u_2, \dots, u_d] = F_{X_1, X_2, \dots, X_d}(x_1, x_2, \dots, x_d) \text{ where } u_i = F_{X_i}(x_i).$$

Copulas provide a way of deconstructing the joint distribution of a set of variables into components (the marginal distributions plus a copula). This means that we can see the nature of the interdependence between the variables.

A copula is determined by the relative order (ranking) of the observations rather than by the exact shape of the marginal distribution.

A copula has three properties:

1. It must be an increasing function of its inputs.
2. If the values of all but one of the marginal CDFs are equal to 1, then the copula is equal to the value of the remaining marginal CDF.
3. The copula must always return a valid probability.

Sklar's theorem

Sklar's theorem says that if F is a joint CDF and $F_1, \dots, F_d$ are marginal CDFs, then there exists a copula C , such that for all $x_1, \dots, x_d \in [-\infty, \infty]$:

$$F(x_1, \dots, x_d) = C[F_1(x_1), \dots, F_d(x_d)]$$

Furthermore if the marginal distributions are continuous, then the copula is unique.

Fundamental copulas

Fundamental copulas represent the three basic (or fundamental) dependencies that a set of variables can display, namely independence, perfect positive interdependence, and perfect negative interdependence. The copulas are:

Independence (or product) copula

The independence (or product) copula is defined in the bivariate case as:

$$C[u, v] = uv$$

Co-monotonic (or minimum copula)

The co-monotonic copula is defined in the bivariate case as:

$$C[u, v] = \min(u, v)$$

Counter-monotonic (or maximum copula)

The counter-monotonic copula is defined in the bivariate case as:

$$C[u, v] = \max(u + v - 1, 0)$$

The minimum and maximum copulas form the upper and lower bounds for all copulas, called the Fréchet-Höfding bounds.

Explicit copulas

Explicit copulas have simple closed-form expressions. An important subclass is that of Archimedean copulas. Archimedean copulas take the form:

$$C[u, v] = \psi^{[-1]}(\psi(u) + \psi(v))$$

where $\psi(t)$ is a generator function and $\psi^{[-1]}$ is a pseudo-inverse generator function.

Three examples of Archimedean copulas are: the Gumbel, Clayton and Frank copulas.

Gumbel copula

The Gumbel copula is defined in the bivariate case as:

$$C[u, v] = \exp \left\{ - \left((-\ln u)^\alpha + (-\ln v)^\alpha \right)^{1/\alpha} \right\}$$

The generator function is $\psi(t) = (-\ln t)^\alpha$ where $1 \leq \alpha < \infty$.

Clayton copula

The Clayton copula is defined in the bivariate case as:

$$C[u, v] = \left(u^{-\alpha} + v^{-\alpha} - 1 \right)^{-1/\alpha}$$

The generator function is $\psi(t) = \frac{1}{\alpha} (t^{-\alpha} - 1)$ where $-1 \leq \alpha < \infty$.

Frank copula

The Frank copula is defined in the bivariate case as:

$$C[u, v] = -\frac{1}{\alpha} \ln \left(1 + \frac{(e^{-\alpha u} - 1)(e^{-\alpha v} - 1)}{(e^{-\alpha} - 1)} \right)$$

The generator function is $\psi(t) = -\ln \left(\frac{e^{-\alpha t} - 1}{e^{-\alpha} - 1} \right)$ where $-\infty < \alpha < \infty$.

Implicit copulas

Implicit copulas are based on well-known multivariate distributions, but no simple closed-form expression exists for them. Examples are the Gaussian copula and the Student's *t*-copula.

Gaussian copula

The Gaussian copula is defined in the bivariate case as:

$$C[u, v] = \Phi_\rho \left[\Phi^{-1}(u), \Phi^{-1}(v) \right]$$

where Φ is the distribution function of the standard normal distribution and Φ_ρ is the distribution function of a bivariate standard normal distribution with correlation ρ .

Combining marginal normal random variables using the Gaussian copula results in the multivariate normal distribution.

The independence, co-monotonic and counter-monotonic copulas are special cases of the Gaussian copula where $\rho=0$, $\rho=+1$ and $\rho=-1$ respectively.

Student's t copula

The Student's t copula defined by:

$$C[u, v] = t_{\gamma, \rho} \left[t_{\gamma}^{-1}(u), t_{\gamma}^{-1}(v) \right]$$

where t_{γ} is the distribution function of a random variable with a Student's t distribution with γ degrees of freedom and $t_{\gamma, \rho}$ is the distribution function of a bivariate Student's t distribution with correlation ρ .

The Student's t copula allows for more flexibility than the normal copula since it involves an extra parameter, namely the number of degrees of freedom, γ .

The Gaussian copula is the limiting case of the Student's t copula as the number of degrees of freedom tends to infinity.

Tail dependence of copula functions

Tail dependence looks at the relationship between two random variables at the extremes of the distributions, ie in the upper and lower tails. Tail dependence can be measured using the coefficients of lower and upper tail dependence, which can be calculated directly from the copula function.

The coefficient of lower tail dependence is defined as:

$$\lambda_L = \lim_{u \rightarrow 0^+} P\left(X \leq F_X^{-1}(u) \mid Y \leq F_Y^{-1}(u)\right) = \lim_{u \rightarrow 0^+} \frac{C[u, u]}{u}$$

The coefficient of upper tail dependence is defined as:

$$\lambda_U = \lim_{u \rightarrow 1^-} P\left(X > F_X^{-1}(u) \mid Y > F_Y^{-1}(u)\right) = \lim_{u \rightarrow 1^-} \frac{1 - 2u + C[u, u]}{1 - u}$$

or as:

$$\lambda_U = \lim_{u \rightarrow 1^-} \frac{\bar{C}[1-u, 1-u]}{1-u} = \lim_{u \rightarrow 0^+} \frac{\bar{C}[u, u]}{u}$$

where $\bar{C}$ is the survival copula, defined as:

$$\bar{C}[1-u, 1-v] = 1 - u - v + C[u, v]$$

For the copulas discussed in this chapter, the table below summarises the coefficients of lower and upper tail dependency.

<i>Copula name</i>	${}_L\lambda$	${}_U\lambda$
Independence	0	
Co-monotonic	1	
Counter-monotonic	0	
Gumbel	0	$2 - 2^{1/\alpha}$
Clayton	$2^{-1/\alpha}$ if $\alpha > 0$ 0 if $\alpha \leq 0$	0
Frank	0	
Gaussian	0 if $\rho < 1$ 1 if $\rho = 1$	
Student's t	> 0 if $\gamma < \infty$, increasing as γ decreases 0 if $\gamma = \infty$ and $\rho \neq 1$ 1 for all γ when $\rho = 1$	

The degree of concordance and the level of tail dependencies exhibited by a particular set of data helps to indicate which copula(s) might be appropriate to consider using.

The practice questions start on the next page so that you can keep the chapter summaries together for revision purposes.



Chapter 17 Practice Questions

17.1 List, in words, the three technical properties which a copula function must satisfy to ensure that it correctly captures the properties expected of a joint distribution function.

17.2 An investor purchases three 5-year bonds from different companies within the same industry sector. The probability that an individual bond defaults within the first year is 10%.

Exam style

(i) Using a Gumbel copula with parameter $\alpha = 2$, calculate the probability that all three bonds default within the first year. [3]

(ii) Discuss the suitability of the Gumbel copula in this situation. [3]

[Total 6]

17.3 For the Clayton copula:

Exam style

(i) Determine whether the generator function $\psi(t) = \frac{1}{\alpha}(t^{-\alpha} - 1)$ is valid. [3]

(ii) Determine the inverse generator function. [1]

(iii) Derive the Clayton copula function in the bivariate case. [3]

[Total 7]

17.4 For the Frank copula:

Exam style

(i) Determine whether the generator function $\psi(t) = -\ln\left(\frac{e^{-\alpha t} - 1}{e^{-\alpha} - 1}\right)$ is valid. [3]

(ii) Determine the inverse generator function. [1]

(iii) Derive the Frank copula function. [3]

[Total 7]

17.5 (i) Derive the coefficient of lower and upper tail dependence for the Clayton copula in the case where the parameter $\alpha > 0$. [4]

Exam style

(ii) Comment on how the value of the parameter α affects the degree of lower tail dependence in the case of the Clayton copula. [1]

[Total 5]

17.6 Derive the coefficient of lower tail dependence for the Gumbel copula in the case where the parameter $\alpha > 0$. [4]

Exam style

- 17.7 Let X and Y be two random variables representing the future lifetimes of two 40-year old individuals. The two lives are married.

Exam style

You are given that:

$$P(X \leq 25) = 0.17831 \text{ and } P(Y \leq 25) = 0.11086$$

- (i) Calculate the joint probability that both lives will die by the age of 65 using:
- (a) the Gumbel copula with $\alpha = 5$
 - (b) the Clayton copula with $\alpha = 5$
 - (c) the Frank copula with $\alpha = 5$. [6]
- (ii) Comment on the results as well as on which copula you think is most appropriate to use for modelling joint life expectancy. [3]
- [Total 9]

- 17.8 You are considering whether there is a link between heights and weights, and have gathered some pairs of data:

$(172\text{cm}, 68\text{kg}), (182\text{cm}, 70\text{kg}), (158\text{cm}, 75\text{kg}), (150\text{cm}, 60\text{kg}), (174\text{cm}, 65\text{kg})$

Calculate Spearman's rho for this dataset.



Chapter 17 Solutions

17.1 Three technical properties a copula function must satisfy are:

1. A copula is an increasing function of its inputs.
2. If all the marginal CDFs are equal to 1 except for one of the marginal CDFs then the copula function is equal to the value of that one marginal CDF.
3. A copula function always returns a valid probability.

17.2 (i) **Calculation of probability all three bonds default within the next year.**

Let T_i be the time until default of bond i where $i = 1, 2, 3$. We want to calculate the joint probability:

$$P(T_1 \leq 1, T_2 \leq 1, T_3 \leq 1) = C[u_1, u_2, u_3]$$

where $u_i = P[T_i \leq 1] = 0.1$ for $i = 1, 2, 3$. [1]

Using the Gumbel copula with parameter $\alpha = 2$, we have:

$$\begin{aligned} P(T_1 \leq 1, T_2 \leq 1, T_3 \leq 1) &= C[u_1, u_2, u_3] \\ &= \exp \left\{ - \left((-\ln u_1)^2 + (-\ln u_2)^2 + (-\ln u_3)^2 \right)^{1/2} \right\} \\ &= \exp \left\{ - \left(3(-\ln 0.1)^2 \right)^{1/2} \right\} \\ &= 0.0185 \text{ or } 1.85\% \end{aligned} \quad [2]$$

(ii) **Suitability of the Gumbel copula**

The Gumbel copula exhibits (non-zero) upper-tail dependence, the degree of which can be varied by adjusting the single parameter. However, it exhibits no lower tail dependence. [1]

Hence, the Gumbel copula is appropriate if we believe that the three investments are likely to behave similarly as the term approaches five years but not at early durations. [½]

This is unlikely to be the case though. If one bond defaults early on, then it may be indicative of problems in the industry sector or the economy and so the other investments may also be likely to default early on. [½]

If we believe the performance of investments issued by companies within the industry are much more closely associated (eg subject to the same systemic and operational risk factors), then a copula that exhibits both lower and upper tail dependence, such as the Student's t copula, may be more appropriate. [1]

17.3 (i) Is the Clayton generator function valid?

- $\psi(0) = \lim_{t \rightarrow 0} \frac{1}{\alpha} (t^{-\alpha} - 1) = \frac{1}{\alpha} \lim_{t \rightarrow 0} \left(\frac{1}{t^{-\alpha}} - 1 \right) = \infty$ [1]

- $\psi(1) = \frac{1}{\alpha} (1^{-\alpha} - 1) = 0$ [1]

- In the range $0 < t < 1$, $t^{-\alpha}$ takes increasing values (starting at 1), so that $t^{-\alpha}$ takes decreasing values, and hence so does $\psi(t) = \frac{1}{\alpha} (t^{-\alpha} - 1)$. [1]

(ii) Inverse generator function

The inverse function is found by rearranging the equation $x = \frac{1}{\alpha} (t^{-\alpha} - 1)$:

$$x = \psi(t) = \frac{1}{\alpha} (t^{-\alpha} - 1) \Rightarrow t = \psi^{-1}(x) = (1 + \alpha x)^{-1/\alpha}$$
 [1]

(iii) Derive the Clayton copula function

$$\begin{aligned} C[u, v] &= \psi^{-1} \left[\frac{1}{\alpha} (u^{-\alpha} - 1) + \frac{1}{\alpha} (v^{-\alpha} - 1) \right] \\ &= \left[1 + \alpha \left(\frac{1}{\alpha} (u^{-\alpha} - 1) + \frac{1}{\alpha} (v^{-\alpha} - 1) \right) \right]^{-1/\alpha} \\ &= \left[1 + (u^{-\alpha} - 1) + (v^{-\alpha} - 1) \right]^{-1/\alpha} \\ &= (u^{-\alpha} + v^{-\alpha} - 1)^{-1/\alpha} \end{aligned}$$
 [3]

17.4 (i) Is the Frank generator function valid?

- $\psi(0) = \lim_{t \rightarrow 0} \left[-\ln \left(\frac{e^{-\alpha t} - 1}{e^{-\alpha} - 1} \right) \right] = \infty$ [1]

- $\psi(1) = -\ln \left(\frac{e^{-\alpha} - 1}{e^{-\alpha} - 1} \right) = 0$ [1]

- In the range $0 < t < 1$, $\frac{e^{-\alpha t} - 1}{e^{-\alpha} - 1}$ takes increasing values, so that $\ln \left(\frac{e^{-\alpha t} - 1}{e^{-\alpha} - 1} \right)$ takes increasing values, and $\psi(t) = -\ln \left(\frac{e^{-\alpha t} - 1}{e^{-\alpha} - 1} \right)$ takes decreasing values. [1]

(ii) **Inverse generator function**

The inverse function is found by rearranging the equation $x = -\ln\left(\frac{e^{-\alpha t} - 1}{e^{-\alpha} - 1}\right)$:

$$x = \psi(t) = -\ln\left(\frac{e^{-\alpha t} - 1}{e^{-\alpha} - 1}\right) \Rightarrow t = \psi^{-1}(x) = -\frac{1}{\alpha} \ln\left[1 + (e^{-\alpha} - 1)e^{-x}\right] \quad [1]$$

(iii) **Derive the Frank copula function**

$$\begin{aligned} C[u,v] &= \psi^{-1}\left[\left(-\ln\left(\frac{e^{-\alpha u} - 1}{e^{-\alpha} - 1}\right)\right) + \left(-\ln\left(\frac{e^{-\alpha v} - 1}{e^{-\alpha} - 1}\right)\right)\right] \\ &= \psi^{-1}\left[-\ln\left(\left(\frac{e^{-\alpha u} - 1}{e^{-\alpha} - 1}\right)\left(\frac{e^{-\alpha v} - 1}{e^{-\alpha} - 1}\right)\right)\right] \\ &= -\frac{1}{\alpha} \ln\left[1 + (e^{-\alpha} - 1)\left(\frac{e^{-\alpha u} - 1}{e^{-\alpha} - 1}\right)\left(\frac{e^{-\alpha v} - 1}{e^{-\alpha} - 1}\right)\right] \\ &= -\frac{1}{\alpha} \ln\left[1 + \frac{(e^{-\alpha u} - 1)(e^{-\alpha v} - 1)}{(e^{-\alpha} - 1)}\right] \end{aligned} \quad [3]$$

17.5 (i)(a) **Coefficients of lower and upper tail dependence – Clayton copula**

The coefficient of lower tail dependence is defined as:

$$\lambda_L = \lim_{u \rightarrow 0^+} \frac{C[u,u]}{u} \quad [1/2]$$

Substituting in for the Clayton copula formula:

$$\begin{aligned} \lambda_L &= \lim_{u \rightarrow 0^+} \left[\frac{(u^{-\alpha} + u^{-\alpha} - 1)^{-1/\alpha}}{u} \right] \\ &= \lim_{u \rightarrow 0^+} \left[\frac{(2u^{-\alpha} - 1)^{-1/\alpha}}{u} \right] \\ &= \lim_{u \rightarrow 0^+} \left[(2u^{-\alpha} - 1)^{-1/\alpha} (u^\alpha)^{-1/\alpha} \right] \\ &= \lim_{u \rightarrow 0^+} \left[(2 - u^\alpha)^{-1/\alpha} \right] = 2^{-1/\alpha} \end{aligned} \quad [1\frac{1}{2}]$$

The coefficient of upper tail dependence is given by:

$$\lambda_U = \lim_{u \rightarrow 1^-} \frac{1 - 2u + C[u, u]}{1 - u} \quad [1/2]$$

From the calculation of the coefficient of lower tail dependence above, we have:

$$C[u, u] = (2u^{-\alpha} - 1)^{-1/\alpha}$$

Therefore:

$$\lambda_U = \lim_{u \rightarrow 1^-} \frac{1 - 2u + (2u^{-\alpha} - 1)^{-1/\alpha}}{1 - u} \quad [1/2]$$

In the limit this fraction has the form $\frac{0}{0}$, which is undefined. We can use L'Hôpital's rule,

$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$, to find the value of the limit:

$$\begin{aligned} \lambda_U &= \lim_{u \rightarrow 1^-} \frac{1 - 2u + (2u^{-\alpha} - 1)^{-1/\alpha}}{1 - u} \\ &= \lim_{u \rightarrow 1^-} \frac{-2 - \frac{1}{\alpha} (2u^{-\alpha} - 1)^{-\frac{1}{\alpha} - 1} (-2\alpha u^{-\alpha - 1})}{-1} \\ &= \lim_{u \rightarrow 1^-} \frac{-2 + (2u^{-\alpha} - 1)^{-\frac{1}{\alpha} - 1} (2u^{-\alpha - 1})}{-1} = 0 \end{aligned} \quad [1]$$

(ii) **Comment**

As α increases, $-\frac{1}{\alpha}$ increases and hence $2^{-1/\alpha}$ increases. So the higher the value of the parameter α , the higher the degree of upper tail dependence. [1]

17.6 Lower tail dependence of the Gumbel copula

The Gumbel copula is expressed as:

$$C[u, v] = \exp \left\{ - \left((-\ln u)^\alpha + (-\ln v)^\alpha \right)^{1/\alpha} \right\} \quad [1]$$

The coefficient of lower tail dependence is given by:

$$\begin{aligned} \lambda_L &= \lim_{u \rightarrow 0^+} \frac{C[u, u]}{u} \\ &= \lim_{u \rightarrow 0^+} \left[\frac{\exp \left\{ - \left((-\ln u)^\alpha + (-\ln u)^\alpha \right)^{1/\alpha} \right\}}{u} \right] \\ &= \lim_{u \rightarrow 0^+} \left[\frac{\exp \left\{ - \left(2^{1/\alpha} (-\ln u) \right) \right\}}{u} \right] \\ &= \lim_{u \rightarrow 0^+} \left[\frac{\exp \left\{ \left(2^{1/\alpha} \ln u \right) \right\}}{u} \right] \\ &= \lim_{u \rightarrow 0^+} \left[\frac{u^{2^{1/\alpha}}}{u} \right] = \lim_{u \rightarrow 0^+} u^{2^{1/\alpha}-1} = 0 \end{aligned} \quad [3]$$

17.7 Calculating probabilities using explicit copulas

We have $u = P(X \leq 25) = 0.17831$ and $v = P(Y \leq 25) = 0.11086$.

(i)(a) Gumbel copula

The joint probability that both lives die by age 65 is given by the Gumbel copula with $\alpha = 5$ as:

$$\begin{aligned} P(X \leq 25, Y \leq 25) &= C[u, v] \\ &= \exp \left\{ - \left((-\ln u)^\alpha + (-\ln v)^\alpha \right)^{1/\alpha} \right\} \\ &= \exp \left\{ - \left((-\ln 0.17831)^5 + (-\ln 0.11086)^5 \right)^{1/5} \right\} \\ &= 0.0986 \end{aligned} \quad [2]$$

(i)(b) **Clayton copula**

The joint probability that both lives die by age 65 is given by the Clayton copula with $\alpha = 5$ as:

$$\begin{aligned}
 P(X \leq 25, Y \leq 25) &= C[u, v] \\
 &= \left(u^{-\alpha} + v^{-\alpha} - 1 \right)^{-1/\alpha} \\
 &= \left(0.17831^{-5} + 0.11086^{-5} - 1 \right)^{-1/5} \\
 &= 0.1089
 \end{aligned}
 \tag{2}$$

(i)(c) **Frank copula**

The joint probability that both lives die by age 65 is given by the Frank copula with $\alpha = 5$ as:

$$\begin{aligned}
 P(X \leq 25, Y \leq 25) &= C[u, v] \\
 &= -\frac{1}{\alpha} \ln \left(1 + \frac{(e^{-\alpha u} - 1)(e^{-\alpha v} - 1)}{(e^{-\alpha} - 1)} \right) \\
 &= -\frac{1}{5} \ln \left(1 + \frac{(e^{-5 \times 0.17831} - 1)(e^{-5 \times 0.11086} - 1)}{(e^{-5} - 1)} \right) \\
 &= 0.0583
 \end{aligned}
 \tag{2}$$

(ii) **Comment**

The Clayton copula gives the highest probability of both lives dying within 25 years. This is because the Clayton copula exhibits lower tail dependence. This means that if one life does not survive for long (*ie* dies early), there is a high probability that the other life will not survive for long (*ie* will also die early). [1]

The Gumbel copula gives the lowest probability of both lives dying within 25 years. This is because the Gumbel copula exhibits upper tail dependence. This means that if one life survives for a long time, there is a high probability that the other life will also survive for a long time. [1]

Studies also suggest that if one member of a married couple dies, this can precipitate the death of the other member ('broken heart syndrome'). On this basis, we might choose to use a copula function where there is a degree of positive interdependence throughout, *eg* the co-monotonic (or minimum) copula. [1]

Although we used the same parameter $\alpha = 5$ in each of the three copula functions, the effect of this parameter on the calculation will vary depending on the copula, and so the results are not directly comparable.

17.8 *Spearman's rho*

<i>Height</i>	<i>Rank</i>	<i>Weight</i>	<i>Rank</i>	d_i	d_i^2
150cm	1	60kg	1	0	0
158cm	2	75kg	5	-3	9
172cm	3	68kg	3	0	0
174cm	4	65kg	2	2	4
182cm	5	70kg	4	1	1

$$s\rho = 1 - \frac{6}{T(T^2 - 1)} \sum_{i=1}^T d_i^2 = 1 - \frac{6}{5(5^2 - 1)} \times 14 = 0.3$$

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